THE SCIENCE BEHIND POPULAR DIETS

KETO DIET

Created in the 1920s for the treatment of epileptic patients, the diet relies on fat as the primary source of fuel for the body. For every three to four portions of fat, the person should consume no more than one portion of carbohydrates and protein combined.

The body goes into a state of ketosis, whereby it starts to burn fat as the primary source of energy, not carbohydrates.

A 2018 research in the *Indian Journal of Medical Research* found a dramatic improvement in weight loss in the short term, but flagged high chances of relapse and nutritional deficiencies if continued over a longer period.

PLANT-BASED DIET

Plant-based diets eliminate all animal products and include vegetables, fruits, nuts, seeds and whole grains. It allows for healthy fats and products with no added sugar.

Studies show that fibre-rich food improves gut bacteria and digestion. This reduces cravings and helps enhance well-being.

A study presented in May 2022 at the European Congress on Obesity found eating vegan for 12 weeks reduced blood sugar in overweight people. Weight loss was also recorded.

HIGH-PROTEIN, HIGH-FIBRE & LOW-CARB DIET

This includes high amounts of protein from animal meat, fish, eggs, legumes and dairy. Very few carbohydrates are allowed. Vegetables high in fibre but less in sugar and fat are allowed. Fatty dairy is also cut down.

Not only does reducing carbohydrates reduce overall calorie intake, but studies show that 20 to 30 per cent of protein calories are burned up in the process of digesting itself. Protein also enhances the appetite-reducing hormone GLP-1 and reduces the hunger hormone Ghrelin. High-fibre vegetables contribute to reduced hunger.

The *American Journal of Clinical Nutrition* notes that a high-protein diet does aid in weight loss. However, there are chances of an increase in uric acid and, consequently, kidney problems if not kept in check.



SHUTTERSTOCK

AYURVEDIC DIET

The diet assumes that each person has a unique body constitution, or 'dosha'. Food is recommended based on individual dosha.

It classifies the body as heat, air or earth, and food is classified accordingly. To balance dosha, one eats certain foods only. For example, eating less of heat-giving food (chillies, meat, fats, garlic) for those who have a heat dosha is said to help restore the health of the body.

There are no large-scale scientific studies to prove whether the diet works

INTERMITTENT FASTING

Intermediate fasters switch between fasting and eating on a regular schedule. The most common practice is to fast for 16 hours and eat for only eight hours a day. Another plan is eating normally for five days but restricting oneself to one 500-600 calorie meal for two days a week.

Restricting calories and not eating for long periods signals the brain to start using stored fat as energy

A 2022 research in the *New England Journal of Medicine* showed that intermediate fasting did lead to reduced weight but it was essentially the same as counting the calories you consume.

NO-ADDED SUGAR DIET

A no-sugar diet restricts all forms of added sugar. This means cutting out not just candy and cake but also other packaged items that have sugar. Some extreme versions also eliminate fruits that have high levels of fructose.

Simple sugar not only spikes insulin levels but also increases the number of calories consumed. On the other hand, complex sugar, which is found in whole grains, is allowed as the fibre takes longer to digest and leads to a gradual release of sugar into the bloodstream. This reduces cravings as well as the overall calorie intake.

An analysis of trials and observational studies published in 2013 in *The BMJ* showed that eliminating added sugar significantly decreased body weight (as long as those calories weren't replaced with others).